

Photo-Synth 2

Play the colours of four photographs.

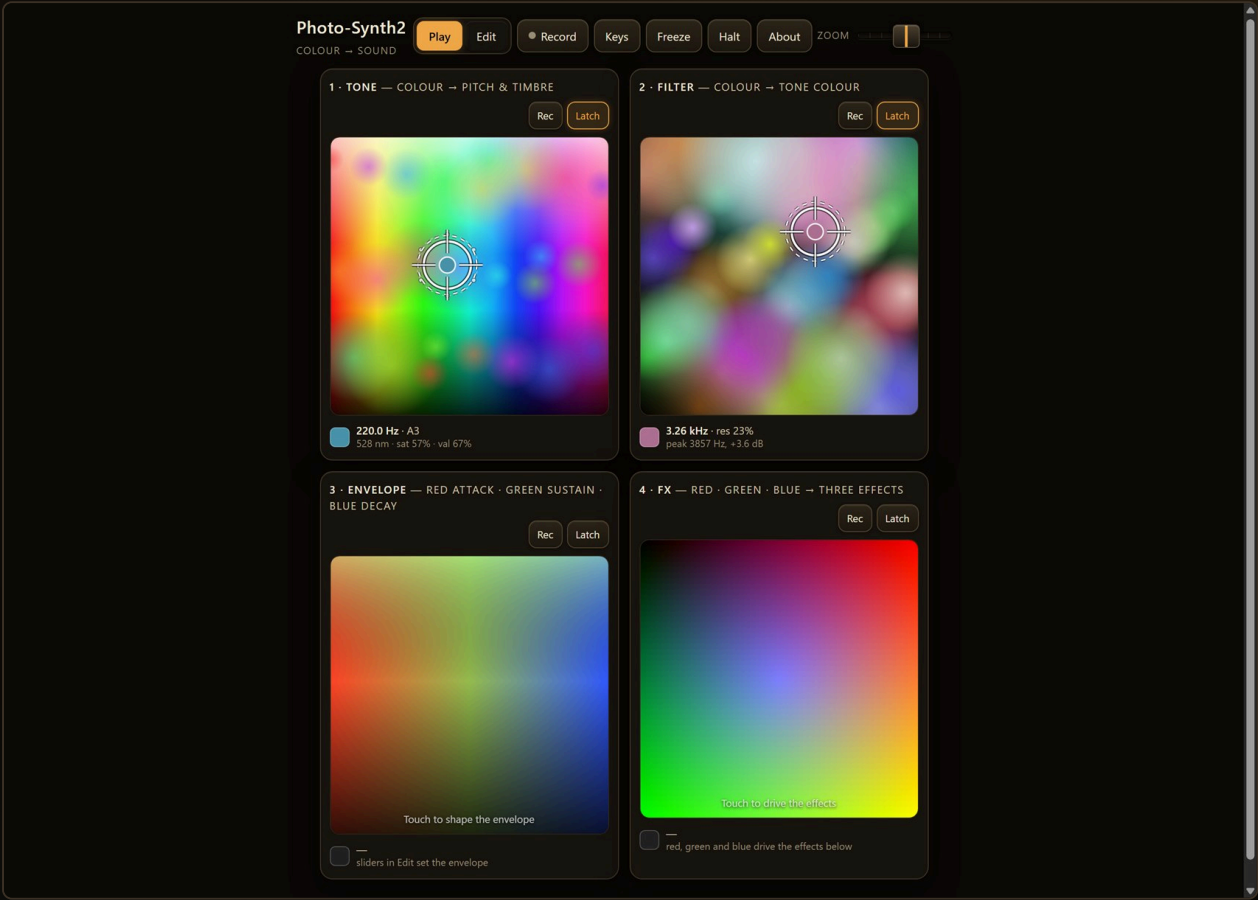


Photo-Synth 2 turns photographs into a playable synthesiser. You do not edit sound with numbers; you drag a finger — or a mouse — across a picture, and the colour under the cursor becomes the sound.

There are four photo pads, and each one listens to colour in a different way. The **first** is the oscillator: hue chooses the pitch, saturation morphs the waveform, brightness sets the loudness. The **second** is the filter. The **third** is the envelope. The **fourth** drives three effect parameters of your choosing. All four work at once, and each can be latched in place or set looping along a recorded gesture, so the instrument keeps moving on its own while you play something else.

Underneath the pictures is a complete virtual-analog voice — two band-limited oscillators that drift a few cents apart, a zero-delay-feedback ladder filter with a saturating core, a sixteen-voice pool, and a seven-module stereo effects rack you can reorder. All of it runs in native C++ inside your DAW, sample-accurate on host MIDI, and every control is automatable.

How the four pads divide the work

PAD	WHAT THE COLOUR UNDER THE CURSOR DOES
1 · Tone	Hue → pitch (red is your base note, violet the top of the range). Saturation → waveform, from a pale sine to a full saw or pulse. Green → detune and vocal edge. Brightness → loudness.
2 · Filter	Brightness → cut-off, from 80 Hz in the shadows to 14 kHz in the highlights. Saturation → resonance. Hue → a resonant peak swept from 180 Hz to 6 kHz.
3 · Envelope	Red → attack (from a 1.5 s swell down to 2 ms). Green → sustain level. Blue → decay and release (30 ms up to a 4 s tail).
4 · FX	The red, green and blue strengths each drive one effect parameter, chosen in the menus under the pad. Black is all three off, white is all three full.

Above all that sits a shelf of **Spectra** — six characters that do not process the sound so much as re-decide how every voice is built: a drone engine, a liquid swirl, a bit-crushed industrial machine, a cathedral, a haunted tape and a swarm of insects. Arm any combination; each hands back exactly what it changed when you switch it off.

Nothing leaves your machine. Photos are decoded and sampled locally, audio is synthesised in the plugin. There is no network access of any kind.

Installing

- Unzip the download and copy the whole `Photo-Synth2.vst3` folder into `C:\Program Files\Common Files\VST3\`.

- Rescan plugins in your DAW. The instrument appears as **Photo-Synth2** by Brokild.
- Windows 10/11, 64-bit. The editor uses the Microsoft Edge WebView2 runtime, which is already present on current Windows installations.

A new instance opens on **Dreamy Hoover** — a stacked, gliding, pentatonic pad with a cursor already resting on the first photo, so the instrument makes a sound the moment it appears. Play with it, then reach for **Naked Oscillator** in the Presets panel when you want the bare engine with nothing between the photo and your ears.

Load it, drag on the first photo, and you are playing. Everything below the first two steps is refinement.

1 Add the instrument to a track.

It arrives with four demo photos already loaded, so it makes sound immediately — you do not need to import anything to start.

2 Drag slowly across photo 1.

The note follows the colour: warm reds are low, cool blues and violets are high. Lift the cursor and the note releases. This is the whole instrument in one gesture.

3 Rest the cursor somewhere on photo 2.

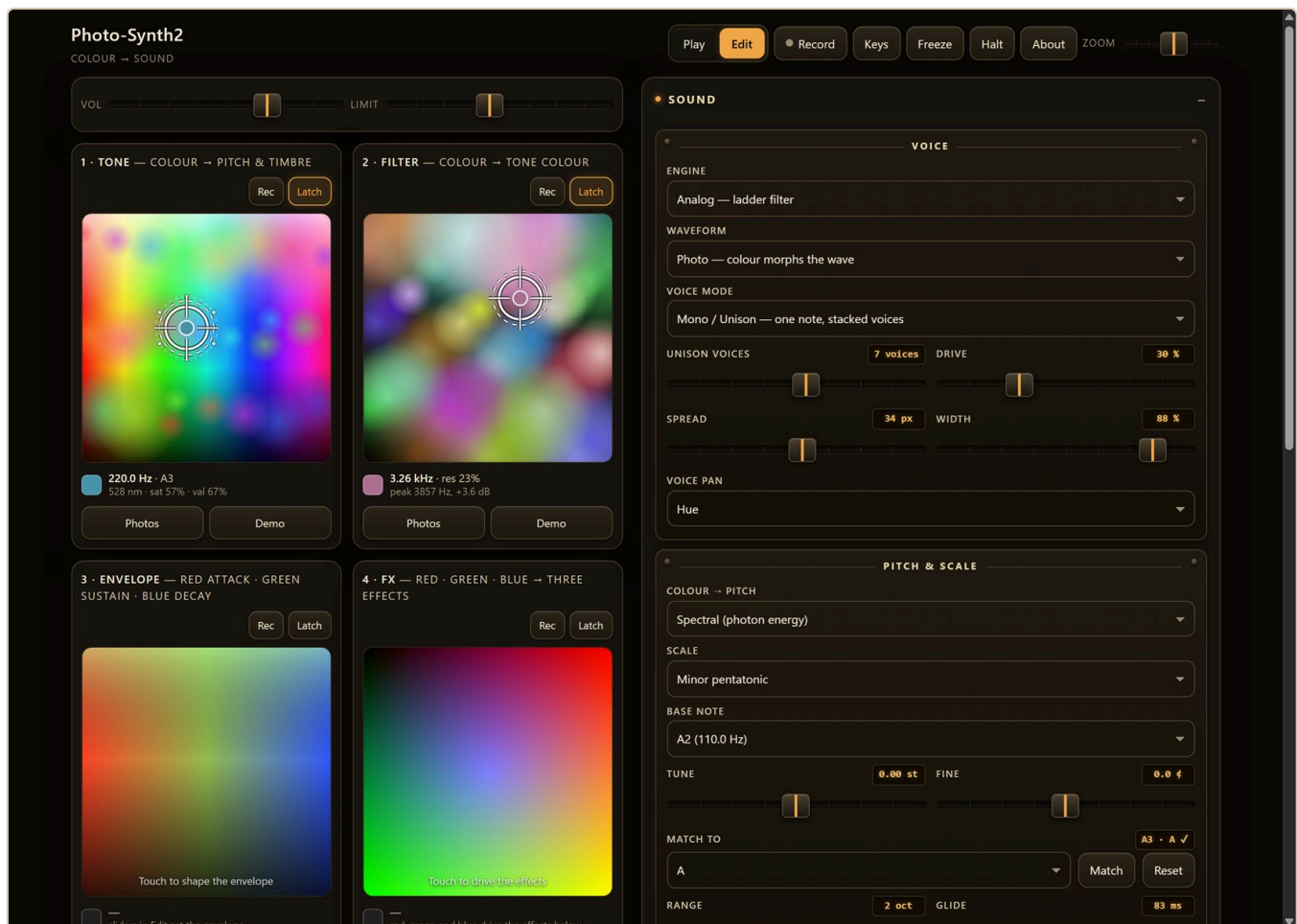
A dark patch mutes the tone, a bright patch opens it up, and a saturated colour adds resonance. Photo 2 keeps whatever colour you left it on — it is a setting, not a gesture.

4 Press *Keys* in the top bar and play the keyboard.

A key sets the **base note**. Everything the photo does is an interval above it, so the same picture can be played in any key. Chords work in Polyphonic voice mode, and MIDI from your DAW plays it too.

5 Switch to *Edit* and open *Sound*.

Here are the voice, the scale, the envelope and the filter type. Then open **Effects** for the pedal rack. Play while you change things; nothing needs to be stopped.



Edit mode. The photos and the analysis strip stay on the left; every panel lives in the right-hand column. **Play** mode hides all of it and leaves only the photos — that is the performance view.

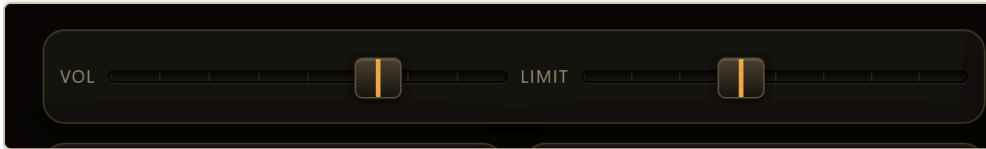
The top bar



From the left: the Play / Edit switch, audio **Record**, the on-screen **Keys**, **Freeze**, **Halt**, **About**, and the interface **Zoom**.

CONTROL	WHAT IT DOES
Play / Edit	Play shows only the four photos. Edit reveals the transport, the analysis strip and all panels.
Record	Captures the plugin's own output. Save WAV in the File panel writes it as 24-bit stereo. Maximum 180 seconds.
Keys	Shows or hides the on-screen keyboard docked at the bottom.
Freeze	Holds the delay tail forever: the loop's input closes, feedback goes to unity under a soft limiter, damping opens wide. Play over the frozen bed.
Halt	A tape stop. Pitch spins down to a halt; press again to spin back up.
Zoom	Scales the whole console uniformly, for large or small plugin windows.

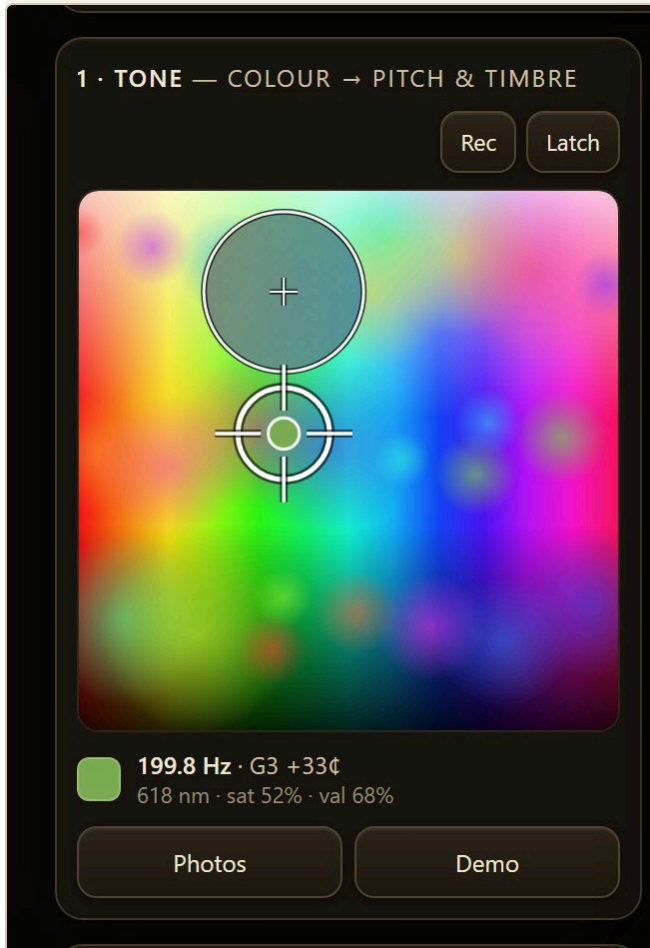
Volume and the limiter



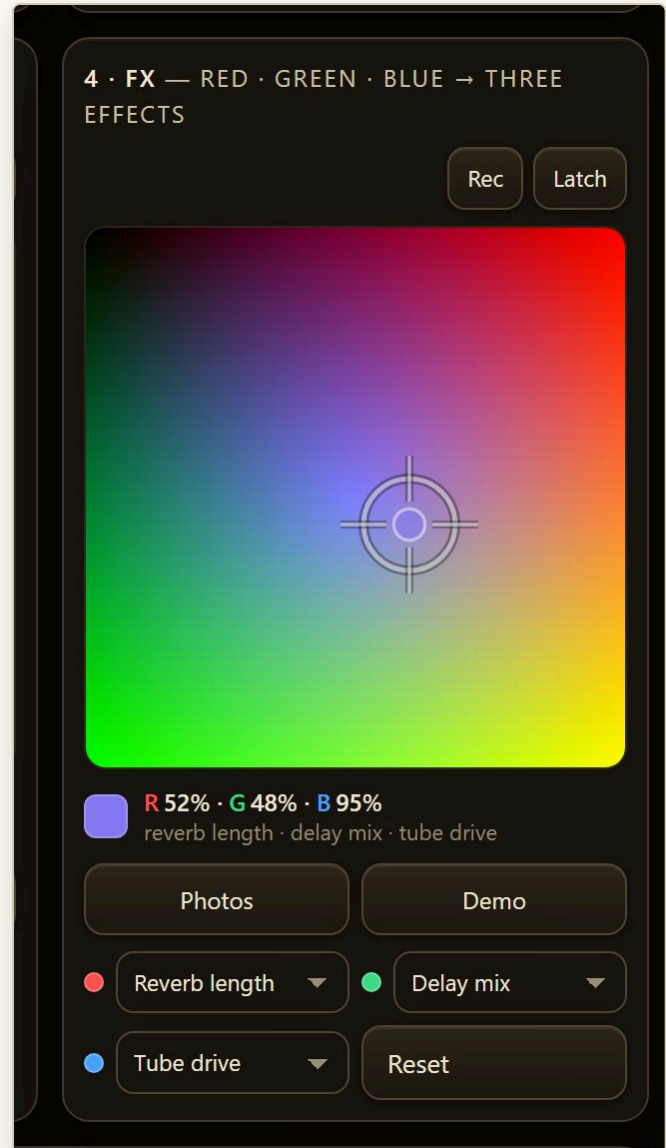
Vol is the master level. **Limit** is a transparent peak limiter.

The limiter is deliberately plain: pure downward gain riding above a threshold, no makeup gain and no tone shaping, so it does not colour the sound. It is there because a resonant ladder filter with the envelope sweeping it can produce sudden peaks. At the default 40 % it catches those without ever touching ordinary playing. Turn it up if you are pushing resonance and drive hard; turn it to zero if you would rather do all the limiting yourself downstream.

Every pad shares the same four controls — **Rec**, **Latch**, **Photos** and **Demo** — and differs only in what it does with the colour it samples.



Pad 1 · Tone. While you drag, a magnifier shows the pixels under the cursor, and the readout gives the exact frequency, the nearest note with its cents offset, the sampled wavelength, saturation and brightness.

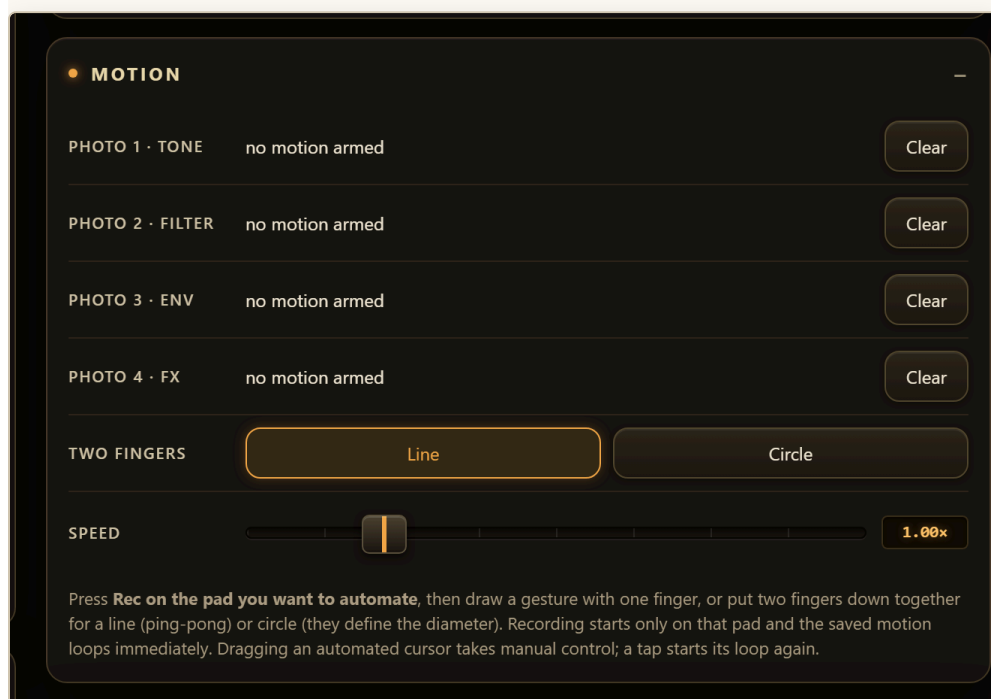


Pad 4 · FX. Three menus assign what red, green and blue each drive. Out of the box: red → reverb length, green → delay mix, blue → tube drive. **Reset** returns those three sliders to their defaults and releases the pad.

Pad controls

CONTROL	WHAT IT DOES
Photos	Loads an image from disk (JPEG, PNG, and anything else Windows can decode). The photo is stored inside the plugin, so it travels with your project and your presets.
Demo	Loads the pad's generated demo image — a hue sweep for Tone, soft colour blobs for Filter, a red-to-blue ramp for Envelope, and an RGB field for FX.
Latch	Leaves the cursor where you lifted it. On Tone the note keeps sounding; on the other pads the setting is held. Tap the pad again to move it, or switch Latch off to release.
Rec	Arms motion recording for that pad alone — see the next page.

Motion: gestures that loop



The Motion panel lists what is armed on each pad, and sets the shape that two fingers describe and the global replay speed.

Press **Rec** on the pad you want to automate, then draw. Release, and the gesture starts looping immediately — the cursor retraces your path, sampling colour as it goes, at the speed set by **Speed**. Instead of drawing you can put **two fingers** down together to define a straight **line** (which ping-pongs) or a **circle** (the two points set the diameter); the choice is the *Two fingers* switch in the Motion panel. Every new note re-phases all running motions, so a phrase lands the same way each time. Dragging an automated cursor takes manual control without erasing the loop; a single tap starts it again. **Clear** discards it.

Live camera is not part of the plugin. A DAW plugin has no business opening your webcam, so the browser version's camera capture and live feed were left out. Everything else from the original is here.

The envelope photo

3 · ENVELOPE — RED ATTACK · GREEN SUSTAIN · BLUE DECAY

Rec

Latch



Touch to shape the envelope



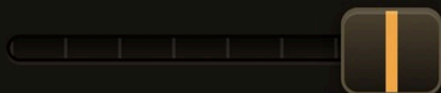
45 ms att · 185 ms dec
sustain 75%

→ AMP

100 %

→ FILTER

28 %



Photos

Demo

Pad 3 · Envelope. Red sets the attack, green the sustain level, blue the decay and release, and the two sliders beneath decide where that shape is sent. Untouched, the fallback sliders in *Sound* → *Envelope* rule the shape instead.

→ **Amp** and → **Filter** appear under this photo, because this is where the envelope is made — and unlike the panels they stay reachable in Play mode. The same pair also appears in *Sound* → *Envelope* beside the fallback shape; they are two faces of one control, so whichever you move, the other follows. → **Amp** is how much of the shape goes to loudness: below 100 % the level blends toward a plain click-free gate, organ-like, so the envelope can be spent on the filter instead. → **Filter** sweeps the cut-off with the envelope, up to five octaves at full — closed when the envelope is down, landing exactly on your filter setting at its peak. Both apply to whichever envelope is in force, the photo's or the sliders'.

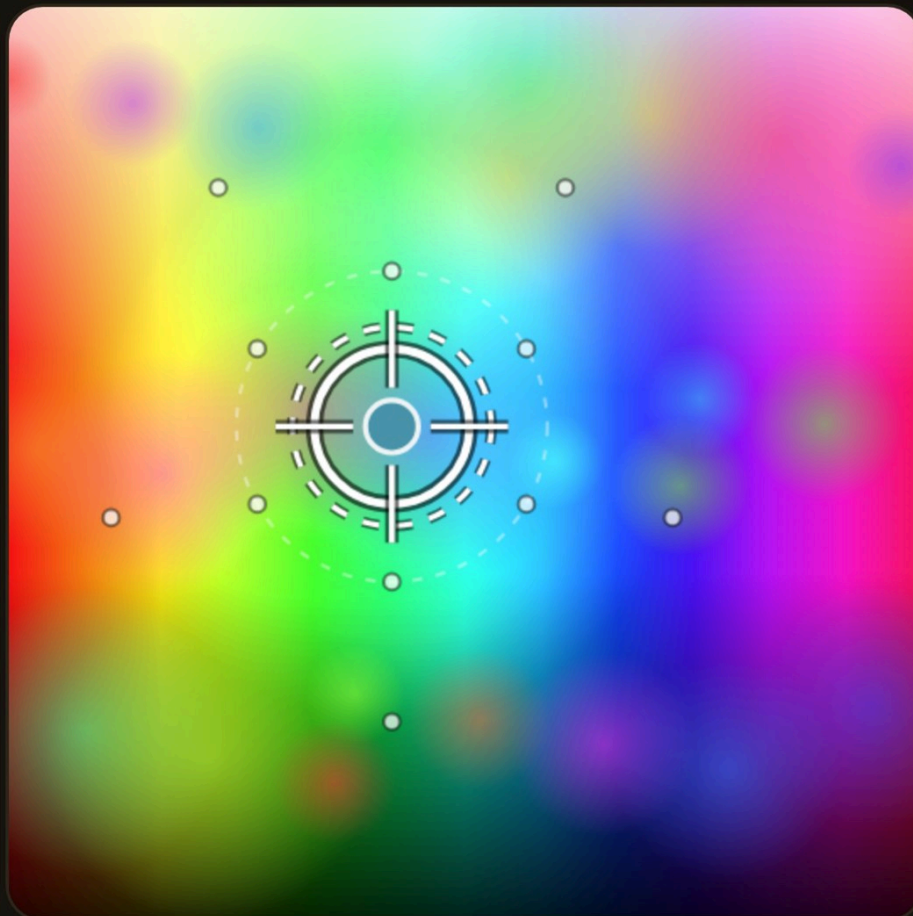
This is the pad that changes how the instrument *feels* rather than how it sounds. A red corner gives you a percussive pluck; a deep blue gives a long ringing tail; a bright green area holds notes up indefinitely. Because it is a photo like any other, you can latch a spot or loop a gesture across it and let the articulation drift while you play.

Seeing what the extra voices read

1 · TONE — COLOUR → PITCH & TIMBRE

Rec

Latch



220.0 Hz · A3
528 nm · sat 57% · val 67%

Photos

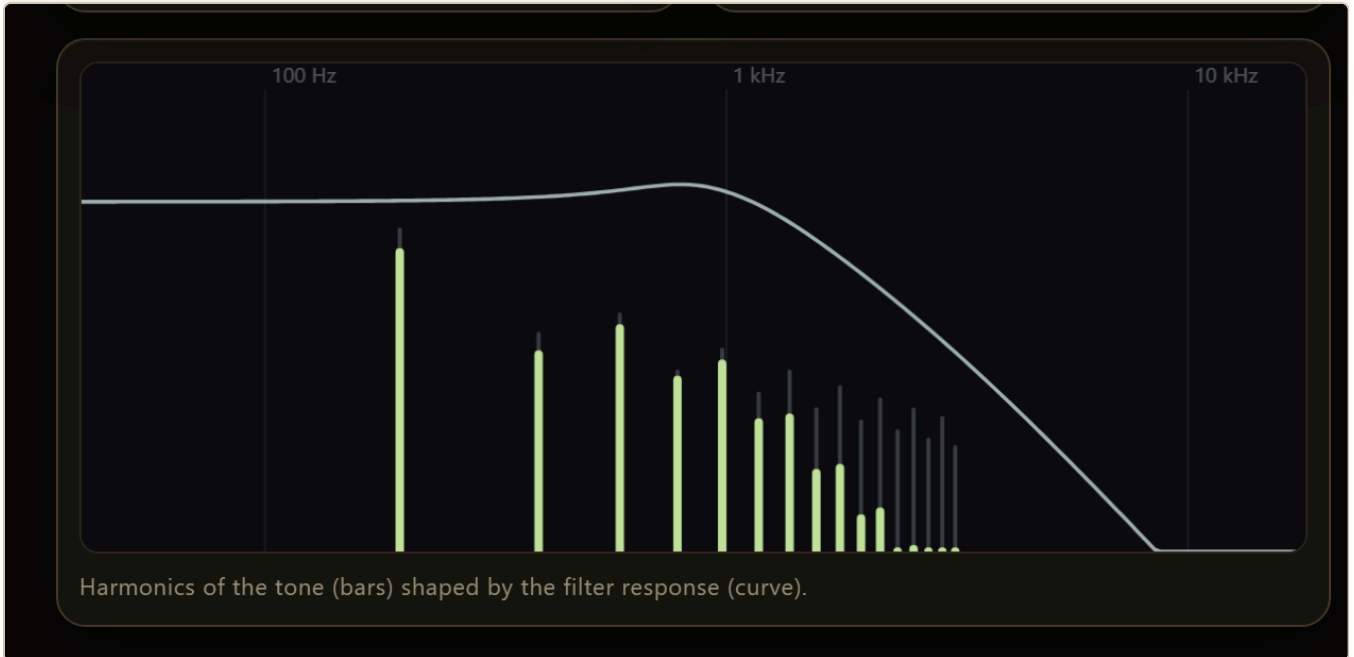
Demo

Twelve voices at a Spread of 46 px: the halo is the patch of photo in play, and every dot is a point that feeds a voice.

Stack more than one voice and they stop reading a single pixel. Voice one plays the colour directly under the cursor; the others sample a ring around it at the **Spread** radius, so the local texture of the photograph becomes a chord. Past seven voices a second, wider ring appears at 1.9× the radius. In **Polyphonic** mode it is each note's oscillators that spread, at 1.0, 1.3 and 1.6× instead.

The tone pad shows you this directly: a soft brightening out to the widest radius in play, a dashed hairline at the Spread radius itself, and a dot on every point that is actually being sampled. Move the Spread slider — now up to 60 px — and you can see the patch of picture the chord is being drawn from. It is the surest way to understand why one part of a photo sounds rich and another sounds flat.

The analysis strip



Bars are the harmonics of the tone you are playing; the curve is the filter your second cursor is describing. Faint bars are the raw harmonic, solid bars are what survives the filter.

Five engraved units: the voice itself, how colour maps to pitch, the envelope, the Dark Drone engine, and the filter type.

VOICE

ENGINE

Analog — ladder filter

WAVEFORM

Photo — colour morphs the wave

VOICE MODE

Mono / Unison — one note, stacked voices

UNISON VOICES

1 voice

DRIVE

25 %

SPREAD

14 px

WIDTH

65 %

VOICE PAN

Hue

PITCH & SCALE

COLOUR → PITCH

Spectral (photon energy)

SCALE

Continuous (no steps)

BASE NOTE

C3 (130.8 Hz)

RANGE

3 oct

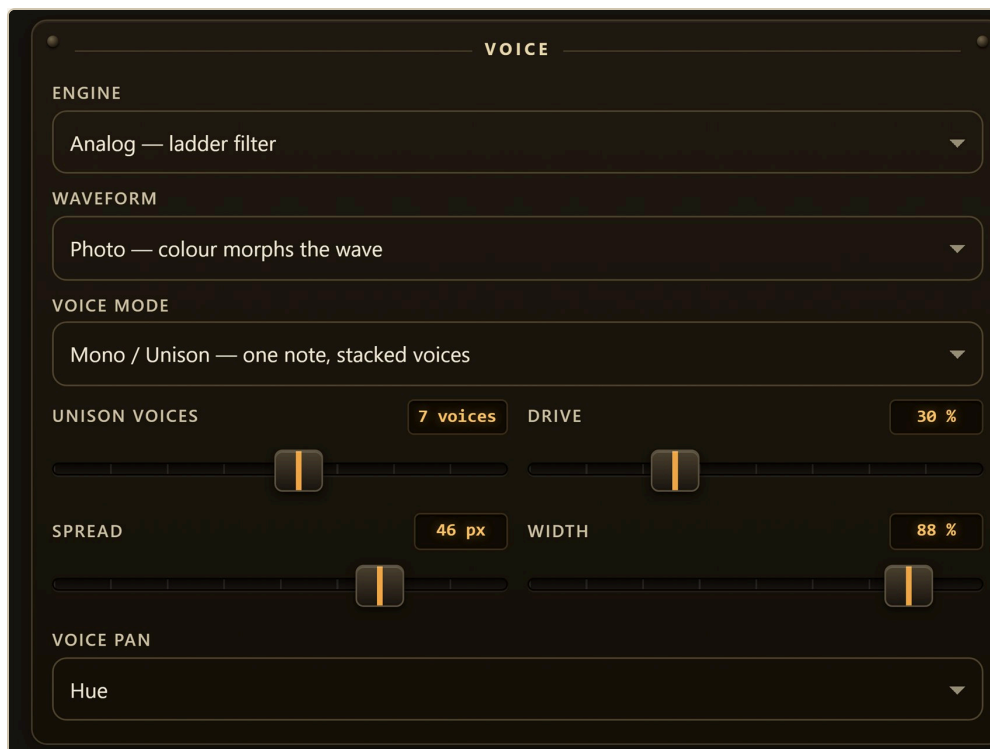
GLIDE

31 ms

The Voice and Pitch & scale units.

Voice

CONTROL	WHAT IT DOES
Engine	Analog is the virtual-analog voice: two band-limited oscillators through a Moog-style ladder filter, running at twice the output rate. Additive is ten pure sine partials and a clean biquad — glassier, simpler.
Waveform	Photo lets the picture morph the wave from sine through triangle to saw or pulse, as saturation rises. The other four — Sine, Triangle, Sawtooth, Square — pin both oscillators to that shape, band-limited and alias-free, and the photo then only affects pitch, loudness and detune.
Voice mode	Mono / Unison plays one note with stacked voices. Polyphonic plays up to four notes at once.
Unison voices	1–12. Extra voices sample the colours in rings around your cursor, so the local texture of the photo becomes a chord riding on one note. Uniform areas thicken into unison; detailed areas scatter into intervals.
Notes / Osc per note	Polyphonic mode only: how many simultaneous notes (1–4) and how many photo-sampling oscillators each note carries (1–4). Oldest note is stolen when you exceed the count.
Drive	How hard the ladder's saturating core is pushed. Thickens rather than distorts — the level is compensated.
Spread	The radius, in pixels, of the ring the extra voices sample from. Small values give a tight unison, large values a wide scatter.
Width	Stereo width. At zero the two channels are identical for true mono compatibility.
Voice pan	Whether voices are panned by the hue they sampled or by their horizontal cursor position .



Voices and **Spread** decide how wide a patch of photo one note reads.

Pitch & scale

CONTROL	WHAT IT DOES
Colour → pitch	Spectral reads hue as a wavelength and follows photon energy, so the intervals feel physical. Hue angle is a plain linear sweep around the colour wheel.
Scale	Continuous, Chromatic, Major, Minor pentatonic, Major pentatonic or Whole tone. Anything but Continuous snaps the colour's interval onto the scale, which makes the pads far easier to play in tune.
Base note	The note a pure red pixel plays. Also set by the keyboard, by your computer keys, and by incoming MIDI.
Range	How many octaves the full colour sweep covers, 1 to 4.
Glide	How long the pitch takes to slide when the colour changes. A little glide makes dragging sound like an instrument; none makes it sound like a scanner.

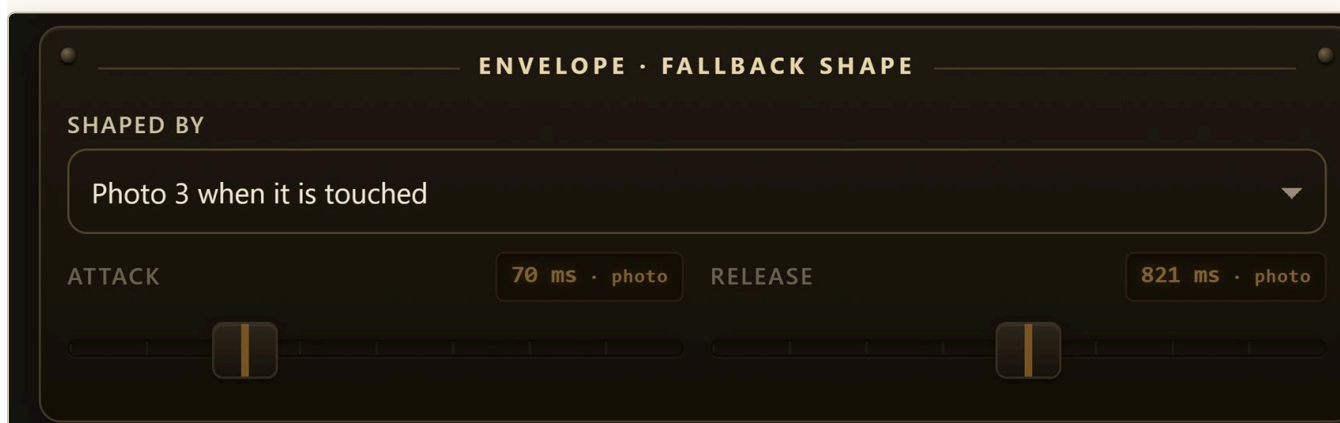


Tuning: **Tune** and **Fine** shift everything the synth plays, and **Match to** is a tuner that does the arithmetic for you.

Tuning the whole instrument

Tune slides continuously over ± 12 semitones — no steps of its own, so it can sit between notes — and **Fine** adds ± 100 cents on top. Both offset every note the instrument plays: the base note, the keyboard, DAW MIDI and an imported MIDI file. They move the base frequency only, so a held note glides to the new pitch rather than retriggering, and you can tune into something while it sounds.

The **Match to** row is a tuner. Its readout is live — the note sounding right now, and how far away the note you picked is, in cents. **Match** then moves Tune and Fine so the sounding note becomes that one, always by the shortest distance: a G matched to A goes up two semitones, never down ten, and never by whole octaves. It reads the frequency the instrument is actually synthesising rather than analysing the output, so it stays exact even while a recorded motion sweeps the pitch around — it catches the note as of the moment you press it. Because the colour under the cursor sets the interval, moving the cursor afterwards moves the pitch again: Match tunes the instrument, it does not pin one note. **Reset** returns both sliders to concert pitch.



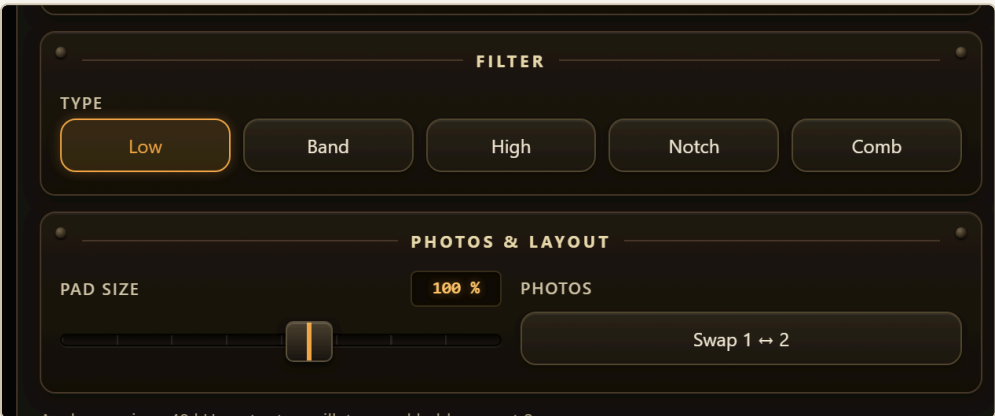
The **Envelope** unit. The drone engine that used to sit beside it now lives in **Spectra**, as Dark Drone Spectrum.

Envelope • fallback shape

Photo 3 takes the envelope over the moment you touch it, and from then on these two sliders are the fallback. **Shaped by** decides which is in charge: *Photo 3 when it is touched* is the normal behaviour, and *These sliders only* hands control back without having to clear the photo. While the photo is in charge the two sliders grey out and go inert, so you can see at a glance why they are doing nothing.

CONTROL	WHAT IT DOES
Shaped by	Photo 3 when touched, or these sliders only.
Attack / Release	The basic shape, used whenever photo 3 is not in charge. Without the photo the decay is fixed at 250 ms and the sustain is full, so this is a plain attack-and-release envelope.
→ Amp / → Filter	The same routing that sits under photo 3, repeated here beside the shape. Move either copy and the other follows.
Legato	How long the note is held after you lift off photo 1, up to half a second, so tapping from colour to colour glides instead of retriggering every note.

Looking for Dark drone? It has moved, with five new companions, into the **Spectra** panel — see section 05.



Filter type, pad size and the photo swap.

Filter type

TYPE	CHARACTER
Low	The classic 24 dB ladder low-pass. Dark patches of photo 2 mute the tone.
Band	Keeps a band around the cut-off and discards the rest — thin, vocal, focused.
High	Removes everything below the cut-off. Bright patches become airy and thin.
Notch	The inverse of band-pass: sweeps a hollow through the sound instead of a peak.
Comb	A tuned feedback delay whose resonances sit at every multiple of the cut-off. With high resonance it rings like a struck string — superb for metallic drones.



Q factor scales the resonance the photograph asks for.

The photo sets the resonance — saturation in photo 2 pushes it up — and **Q factor** scales what it asks for, from a quarter to four times, with the centre position leaving the original behaviour exactly as it was. It reaches every mode: the ladder's resonance in Low, Band, High and Notch, the comb's internal feedback, and the biquad Q of the additive engine. Turn it up for surgical notches and singing band-passes; turn it down when a bright photograph is making the filter shout.

Photos & layout: *Pad size* changes the aspect of all four pads, and *Swap 1 ↔ 2* exchanges the tone and filter photos.

A Spectrum is a character rather than an effect. It does not sit in the signal path waiting for audio; it changes how every voice is built in the first place — which is why each one wears its own enclosure instead of a shared faceplate, and why none of them costs anything at all while its switch is off.

• SPECTRA

A Spectrum module is a character rather than an effect: it does not sit in the signal path but re-shapes how every voice is built. Arm any combination — where two reach for the same control the most recently armed wins, and each hands back what it took when released.

⌂ DARK DRONE SPECTRUM

+

?



Psychedelic Pink Spectrum

+

?



INDUSTRIAL BLACK SPECTRUM

+

?



GLASS CATHEDRAL SPECTRUM

+

?



TAPE SÉANCE SPECTRUM

+

?



INSECT CHOIR SPECTRUM

+

?

Six characters, one switch each. **Tape Séance** is armed here — the lens lights in the module's own colour.

Arming, combining, releasing

The illuminated rocker arms a character. You may arm **any combination**: they stack, and where two of them reach for the same control the most recently armed one wins. Switching one off unwinds the stack to that point and re-applies whatever was armed after it, so every release hands back exactly what that character changed and nothing else. Your own chain order, power switches and slider positions survive untouched.

The (?) beside each switch unfolds a description of what that character actually does, and opens the controls if they are collapsed. Plates stay tight until you ask.



Tape Séance Spectrum open, with its explanation unfolded by the (?).

Where a character's effects sit

Four of the six switch on effects of their own. The **Chain** menu decides where those effects land in your rack: **Pre** pulls them to the front, so everything downstream is coloured by them — a reverb after Industrial Black hears the crush and the stutter, which is usually what you want. **Post** pushes them to the end, treating the finished sound instead. Dark Drone and Insect Choir add no effects at all — they work entirely on the voices — so their Chain row simply says so.



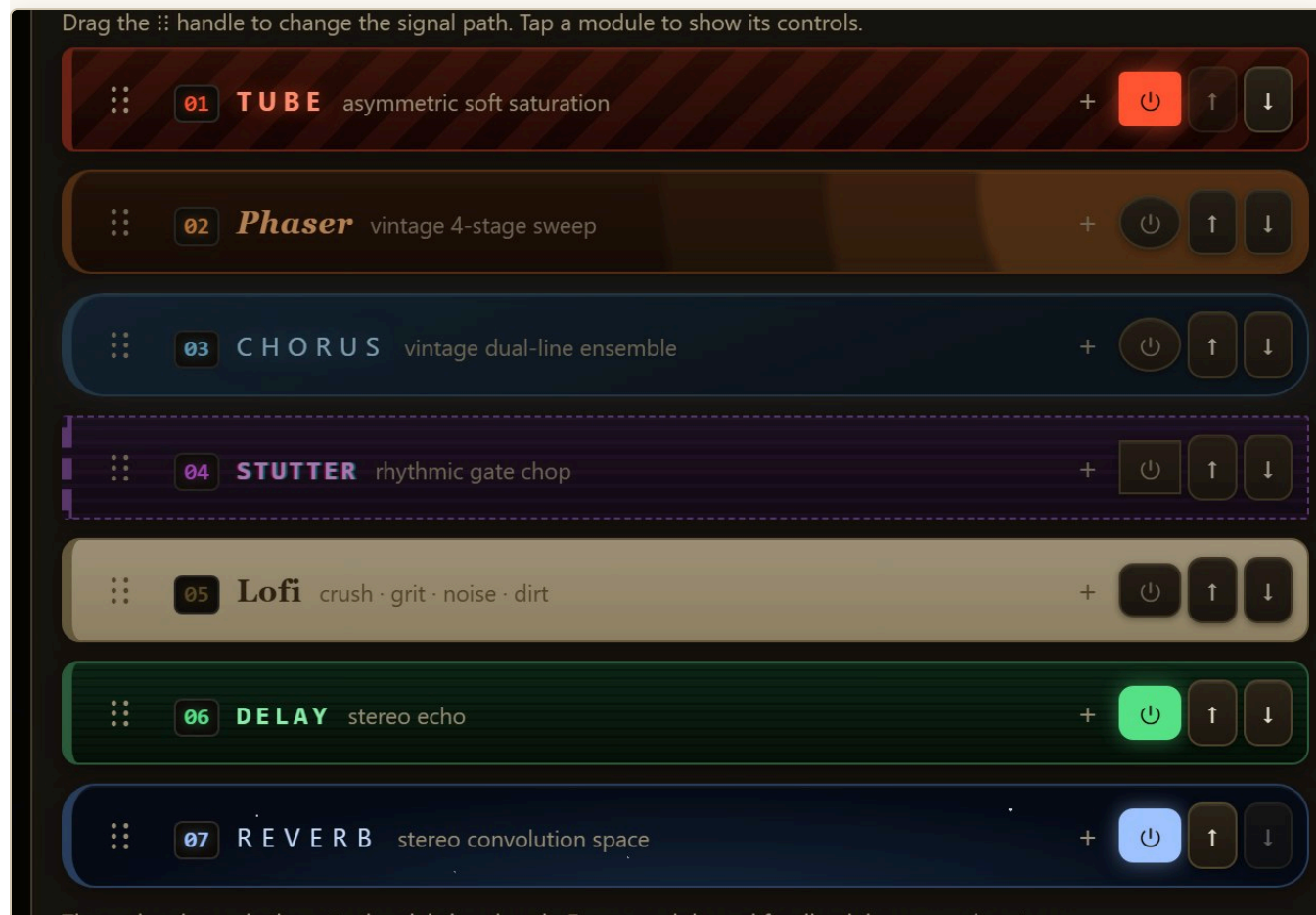
Industrial Black Spectrum, armed. Grind is drive and bit-crush, Chop the gate, Clang the metal.

The six characters

CHARACTER	WHAT IT DOES TO THE INSTRUMENT
Dark Drone	A sub-oscillator under every voice through the same filter, an irregular microtonal cluster instead of tidy unison, an ultra-slow random walk leaning on cut-off, detune and delay time, and notes that sag toward an octave down as they decay. <i>Sub level, Sub octave, Cluster, Env → Pitch, Drift, Drift time.</i>
Psychedelic Pink	Everything smeared and slowly rotating: the stereo image swirls, a slow wave bends the tuning, the filter breathes on its own, and the reverb becomes the reverse character so tails swell in rather than die away. <i>Swirl, Smear, Bloom.</i>
Industrial Black	A square wave into the crusher <i>before</i> the filter, so the ladder has to smooth the wreckage; a comb resonance and a tight metal slapback, a stutter on the gate, and notes that stop dead instead of releasing. <i>Grind, Chop, Clang.</i>
Glass Cathedral	Your own tone sent to church: notes open like light arriving and take their time to leave, the envelope is sent into the filter so every note dawns, and a long hall stands behind it. Everything dirty steps aside while it holds the panel. <i>Air, Halo, Shine.</i>
Tape Séance	A reel that has been in the attic too long: wow and flutter bend the pitch and the echoes together, notes sag flat as they die, and there is dust on everything — hiss, crackle, a dull top end. <i>Wobble, Sag, Dust.</i>
Insect Choir	The voices stop agreeing: they detune into a cluster instead of a chord, each one trembles at its own rate, and small random jumps keep the swarm moving. Arming it raises the voice count. <i>Swarm, Flutter, Skitter.</i>

They are macros, not extra DSP. Every character works by moving the instrument's own controls and by adding a little motion of its own, which is why arming one costs nothing you were not already paying for, why the result is visible on the panels, and why releasing it can put everything back precisely.

Seven stereo modules in a real serial chain. The order shown is the signal path: drag the :: handle, or use the arrows, and the audio graph is rewired under a short master dip so it never pops.



The rack in the **Boutique** skin, where every unit wears its own enclosure. **Tube**, **Delay** and **Reverb** start switched on; the other four start off. A module that is off is removed from the audio graph entirely, so bypassing genuinely saves CPU. The *Skin* menu at the top of the panel switches to **Rack** — uniform faceplates — if you prefer the plainer look.

Tube — asymmetric soft saturation

Drive 0–24 dB, **Tone**. A biased triode-style curve adds gentle even harmonics; four-times oversampled with DC blocking. Loudness-compensated, so driving it thickens the sound without getting louder.

Delay — stereo echo

Mix, **Time** 40–900 ms, **Feedback** up to 112 %, **Offset** L↔R ±250 ms, **Character**. Past unity the loop self-oscillates into a soft limiter instead of blowing up.

Reverb — stereo convolution

Mix, **Character**, **Length** 0.2–6 s. Room, Hall, Plate, **Spring** (a chain of dispersive chirps — the classic boing) and **Reverse**,

Chorus — vintage dual-line ensemble

Mix, **Rate**, **Depth**. Two delay lines modulated in opposite phase, wet path rolled off dark like an old bucket-brigade unit.

Stutter — rhythmic gate chop

Amount, **Rate** 1–16 Hz. A smoothed gate, so the chopping never clicks. Put it *before* delay and reverb and the tails survive the cuts.

Lofi — crush · grit · noise · dirt

Crush (sample-rate reduction and bit grit), **Noise** (hiss plus sparse crackle), **Dirt** (soft normalised saturation). Everything is smoothed per sample, so the FX photo can drive it without clicks.

whose impulse swells in so notes arrive with a drawn-in breath.

Phaser — vintage 4-stage sweep

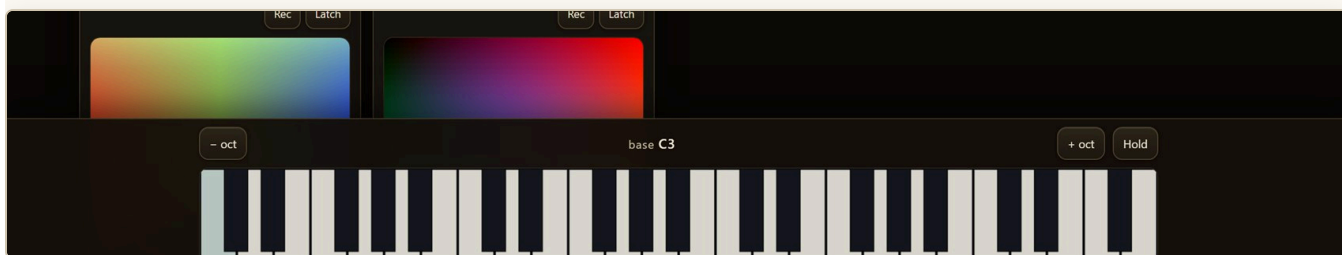
Mix, Rate, Depth. Four swept all-pass stages with a little feedback: the slow hollow swoosh of a seventies stomp box.



Tap a module's name to fold its controls open or shut.

Delay character

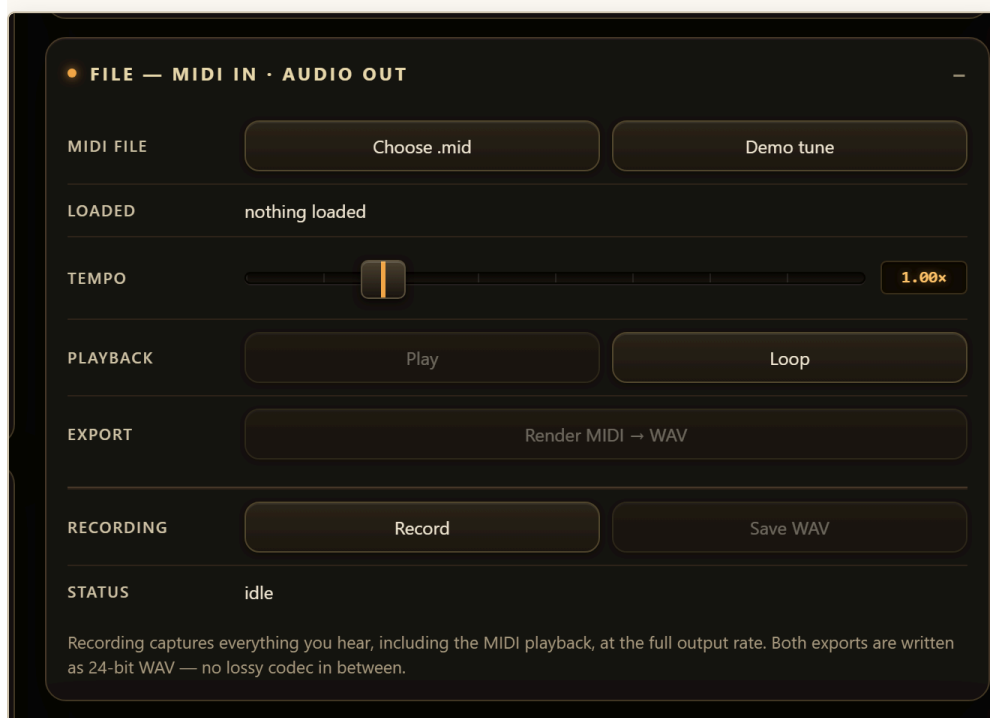
Clean gives digital repeats. **Tape** runs the feedback loop through gentle saturation, darker damping, a low cut and a slow wow, so every repeat gets softer, dirtier and slightly unstable — a worn analog echo. Combined with **Freeze** in the top bar, tape mode degenerates into a beautiful saturating murk you can play over.



The keyboard sets the **base note** — the note a pure red pixel plays. It adapts to the window width, and the teal marker shows the current base.

Use **- oct** and **+ oct** to move the range, and **Hold** to keep the last key down. On a computer keyboard the rows **z s x d c v g b h n j m** and **q 2 w 3 e r 5 t 6 y 7 u** play two octaves from the leftmost key. In Polyphonic voice mode, several keys — or several MIDI notes — sound together as a real chord.

Your DAW plays it too. MIDI from the host is handled natively and sample-accurately, with the same note allocation as the on-screen keyboard. Record a MIDI part on the track and the photos shape every note of it.



MIDI file playback, tempo scaling, offline render and the audio recorder.

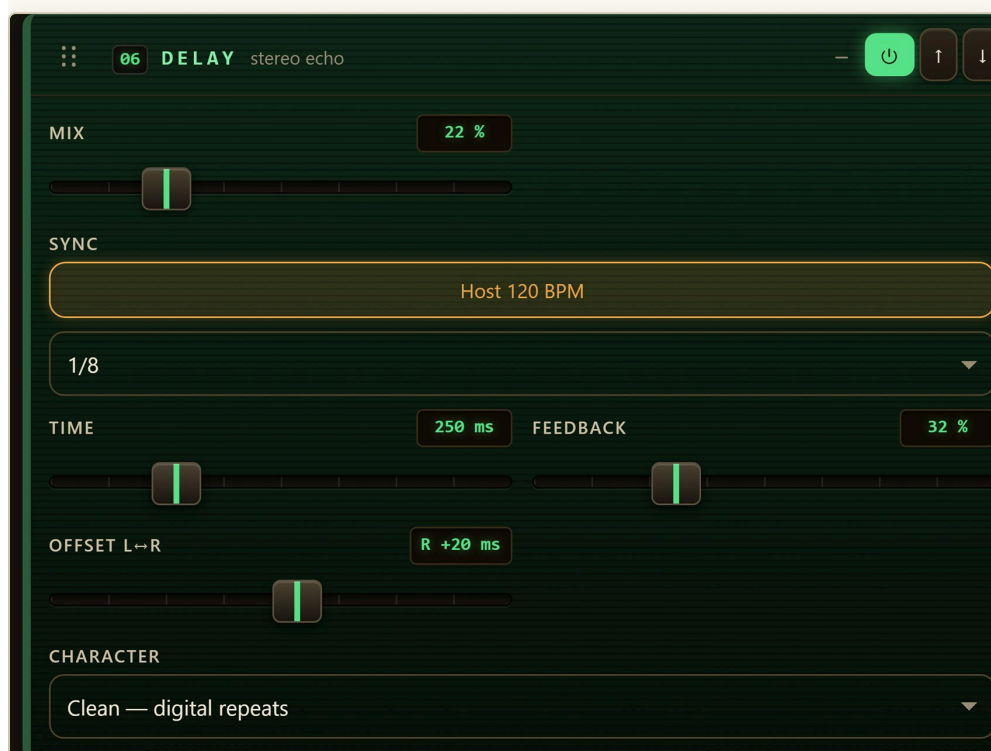
Playing a MIDI file

Choose .mid imports a standard MIDI file, or **Demo tune** loads a short pentatonic phrase to try the path. Its melody drives the base note while the photos keep shaping the sound; the tempo comes from the file and **Tempo** scales it from a quarter speed to four times. In Mono/Unison mode overlapping notes are reduced to the highest one; in Polyphonic mode the file's chords play for real, up to your Notes setting.

Recording and export

ACTION	WHAT HAPPENS
Record	Captures the plugin's output — playing, MIDI, filter sweeps and all — up to 180 seconds.
Save WAV	Writes the capture as 24-bit stereo WAV at the session's sample rate, through a normal Windows save dialog.
Render MIDI → WAV	Renders the loaded MIDI file offline, faster than real time and sample-accurate, at 48 kHz — using the colours currently under the cursors.

Locking the delay and the stutter to the tempo



The **Sync** switch reads the host tempo back to you; the menu beside it picks the note division.

Delay and Stutter can lock to the DAW. Press **Sync**, choose a division — 1/1 down to 1/32, dotted and triplet — and the module's own Time or Rate control is driven from the tempo the host reports, following it when you change it. The control is held while synced so a drag cannot fight it, and the switch reads back the current tempo.

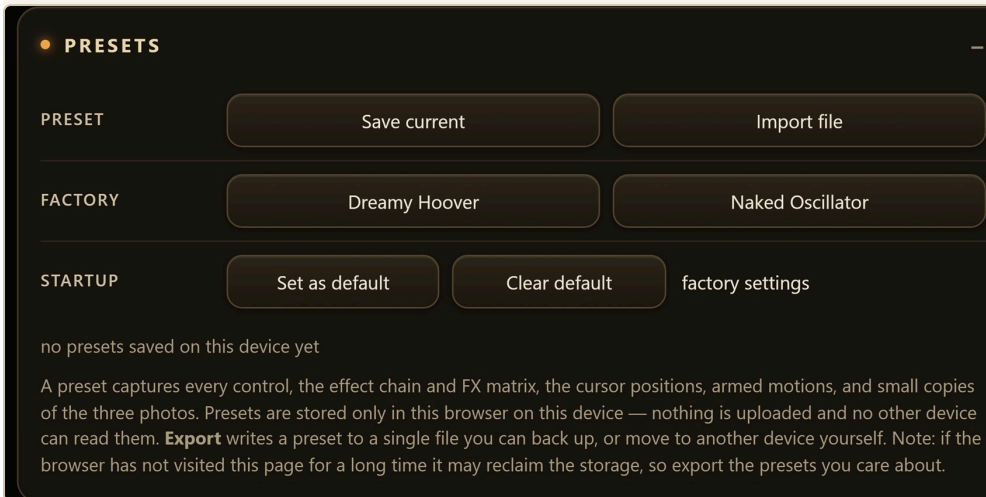
Because the page follows the tempo, a tempo change while the plugin window is closed is not tracked; the last synced value is held until you open it again.

Factory patches and what the instrument opens on

Two patches are built in. **Dreamy Hoover** is what a new instance greets you with: seven voices stacked and gliding, pentatonic so every position on the photograph is musical, chorus and a long hall behind it, and a cursor already resting on the first photo. **Naked Oscillator** is the opposite — one voice, no effects, nothing between the picture and your ears — and it is the honest place to start building a sound of your own. Neither carries photographs, so both dress whatever pictures are loaded.

Once you have a patch you would rather be greeted by, **Set as default** freezes it — controls, chain, armed characters, cursors, motions and small copies of the photographs — and new instances open on that instead. **Clear default** returns to Dreamy Hoover. A patch saved inside a DAW project always wins over both.

Presets



Presets keep everything, photographs included.

A preset captures every control, the effect chain and its power switches, the FX matrix assignments, the cursor positions, the armed motions, and small copies of all four photos. **Save current** stores it on this machine; **Export** writes a single `.photosynth.json` file you can back up or send to someone else; **Import file** reads one back. Your DAW project stores the same thing independently, so reopening a session restores the sound *and* the pictures without any preset management at all.

Nine short experiments, roughly in order of adventurousness. Each one is a handful of moves — the point is to hear what the instrument does, not to build a patch.

1 • The first thirty seconds

Drag slowly across photo 1 in a wide arc, then leave the cursor on a bright saturated area of photo 2. Now drag on photo 1 again.

Everything else in this manual is a variation on this.

2 • Play it in tune

Sound → Pitch & scale: set **Scale** to **Minor pentatonic** and **Range** to **2 oct**. Add a little **Glide**.

Suddenly the photo is an instrument rather than a colour meter — every colour lands on a usable note.

3 • One photo, one chord

Voice mode **Mono / Unison**, **Unison voices** to 8, **Spread** to about 30 px. Put the cursor on a *busy*, detailed part of photo 1 — then on a smooth gradient.

Detail scatters the eight voices into a chord; smooth areas collapse them into a thick unison. The texture of the picture becomes harmony.

4 • Real chords from your DAW

Voice mode **Polyphonic**, **Notes** 4, **Osc / note** 2. Record a slow four-note chord progression on the track and let it play while you move the cursor on photo 1.

The chord holds while the timbre of every note travels with your cursor.

5 • A filter that plays itself

Press **Rec** on photo 2, put two fingers down (or click and drag with the Motion shape set to **Circle**) to describe a circle over a colourful area, and release. Set **Speed** to about 0.5×.

A slow endless filter sweep that follows the picture rather than an LFO — hold a note and listen to it breathe.

6 • Percussive from a photograph

Latch a **deep red** spot on photo 3 (fast attack, short tail), set **Envelope** → **Filter** to about 60 %, filter type **Low**, and play staccato notes on the keyboard.

Plucks and blips, with the filter snapping shut behind each note.

7 • Metallic drone

Filter type **Comb**. Latch photo 1 on a saturated colour and photo 2 on something bright. Engage **Dark Drone**, **Sub level** up, **Cluster** around 30 ¢, **Drift** 60 %, **Drift time** long. Leave it running for a few minutes.

A struck-string resonance that never quite repeats itself and slowly comes apart. This is what the instrument is best at.

8 • Freeze and play over it

Delay **Character** to **Tape**, **Feedback** around 90 %, **Mix** up. Play a phrase, then hit **Freeze** in the top bar and keep playing.

The phrase is held forever as a saturating tape bed while you improvise on top. Press Freeze again to let it decay.

9 • Let the fourth photo run the effects

Assign the FX pad's three menus to **Lofi crush**, **Chorus mix** and **Reverb mix**, switch those three modules on, then press **Rec** on photo 4 and draw a long wandering gesture.

*The effects morph continuously along a path you drew, in combinations no three LFOs would find. Finish with **Halt**.*

Automation

Around forty controls are exposed to the host as automatable parameters, named exactly as they are labelled in the panels — `vol`, `limiter`, `drive`, `attack`, `release`, `envAmp`, `envFlt`, `satGain`, `delayMix`, `delayTime`, `delayFeedback`, `reverbMix`, `reverbLength`, the phaser, chorus, stutter and lofi controls, and the Dark Drone set. Write automation for them as you would for any synth.

Added since version 1.0, and automatable like the rest: `tune` and `fine` (global tuning), `filterQ` (the resonance scaler), and `spread` now reaching 60 px. The Spectra characters, the tempo-sync switches and the tuner's Match button are panel controls rather than parameters — they move the automatable sliders on your behalf, so what they do is recorded in the project and visible on the panels.

What the project remembers

Saving your session stores every control, the effect order and power switches, the FX matrix, the cursor positions, the armed motions and compressed copies of all four photographs. Reopening the session restores the instrument exactly, pictures included, with no external files to keep track of.

Two things worth knowing

- **The window may be left open for gestures.** The engine keeps sounding with the editor closed and host MIDI plays normally, but the parts that are genuinely driven by the interface — looping motions, the Dark Drone drift walk, the FX photo, and re-sampling of the photos — need the plugin window open to run. For a fixed sound, close it freely.
- **Resonance plus drive can peak.** That is the nature of a ladder filter with the envelope sweeping it, and it is why **Limit** exists. It is on at 40 % by default and transparent at that setting.

Photo-Synth 2

COLOUR → SOUND

This VST comes from the curved mind of **Professor Brokild**, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

Cheers, Peter Bøggild, August 2026.

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Close

The **About** button, in the top bar.

Photo-Synth 2 · Brokild Apps · August 2026. Native VST3 and standalone port of the browser instrument Photo Synth. The interface is the original page; the audio engine — voice pool, ladder filter, effects rack, recorder and offline render — is native C++.

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